

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Classified Practice

Circles

Q001 **Written****Find the equation of the circle with centre (3,-2) and radius 5.**

Q002 Written**Find the equation of the circle with centre $(-3,1)$ and radius 2.**

Classified Practice

Circles

Q003 Written

Find the equation of the circle with centre the origin and radius 4.

Q004 Written**Find the equation of the circle with centre (3,4), radius 6.**

Classified Practice

Circles

Q005 Written

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$.

Q006 Written**Find the equation of the circle with centre the origin, radius 5.**

Classified Practice

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Choose the correct option.**

A $(x - 3)^2 + (y + 2)^2 = 25$

B $(x + 3)^2 + (y + 2)^2 = 25$

C $(x - 3)^2 + (y + 2)^2 = 25 + 1$

D 4

Q008 MCQ**Find the equation of the circle with centre (-3,1) and radius 2:****Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 4$

B $(x + 3)^2 + (y + 1)^2 = 4$

C $(x + 3)^2 + (y - 1)^2 = 4 + 1$

D 6

Classified Practice

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Choose the correct option.**

A $x^2 + y^2 = 16 + 1$

B 6

C 7

D $x^2 + y^2 = 16$

Q010 MCQ**Find the equation of the circle with centre (3,4), radius 6:****Choose the correct option.****A** 9

B $(x - 3)^2 + (y - 4)^2 = 36$

C $(x + 3)^2 + (y - 4)^2 = 36$

D $(x - 3)^2 + (y - 4)^2 = 36 + 1$

Classified Practice

Circles

Q011 MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:**Choose the correct option.**

A 10

B $(x + 2)^2 + (y - 6)^2 = 12$

C $(x + 2)^2 + (y + 6)^2 = 12$

D $(x + 2)^2 + (y - 6)^2 = 12 + 1$

Q012 MCQ**Find the equation of the circle with centre the origin, radius 5:****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 10

D 11

Classified Practice

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Q014 Written**Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.**

Classified Practice

Circles

Q015 **Written****Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.**

Q016 Written**Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.**

Classified Practice

Circles

Q017 **Written****Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.**

Q018 Written**Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.**

Classified Practice

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Choose the correct option.**A circle; centre $(3, -1)$, radius 2

B 3

C 4

D 5

Classified Practice

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:

Choose the correct option.

A 7

B 8

C circle; centre (2, 3), radius 3

D 6

Classified Practice

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Choose the correct option.**

- A 10
- B 11
- C 12
- D circle; centre $(2, -4)$, radius 4

Classified Practice

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Choose the correct option.**

A 12

B 13

C circle; centre (2, 0), radius 2

D 11

Classified Practice

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Choose the correct option.**

- A 13
- B 14
- C circle; centre $(-1, 2)$, radius 5
- D circle; centre $(-1, 2)$, radius 10

Classified Practice

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Choose the correct option.**

A 14

B 15

C circle; centre $(-2, 3)$, radius $\sqrt{13}$

D 13

Classified Practice

Circles

Q025 **Challenge**

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Choose the correct option.

A 0

B centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

C centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 + 1$

D centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13 - 1$

Classified Practice

Circles

Q027 **Written****Find the equation of the circle with centre (2,3) passing through (4,6).**

Q028 Written**Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$.**

Classified Practice

Circles

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).

Q030 Written**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$.**

Classified Practice

Circles

Q031 Written

Find the equation of the circle with centre (2,-1) through the origin.

Q032 Written**Find the equation of the circle with centre the origin through (4,-3).**

Classified Practice

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Choose the correct option.**

A $(x - 2)^2 + (y - 3)^2 = 13 + 1$

B 4

C $(x - 2)^2 + (y - 3)^2 = 13$

D $(x + 2)^2 + (y - 3)^2 = 13$

Q034 MCQ**Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$:****Choose the correct option.**

A $(x + 3)^2 + (y - 1)^2 = 58$

B $(x + 3)^2 + (y + 1)^2 = 58$

C $(x + 3)^2 + (y - 1)^2 = 58 + 1$

D 7

Classified Practice

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Choose the correct option.**

A $(x + 1)^2 + (y + 3)^2 = 25$

B $(x - 1)^2 + (y + 3)^2 = 25 + 1$

C 10

D $(x - 1)^2 + (y + 3)^2 = 25$

Q036 MCQ**Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$:****Choose the correct option.**

A $(x + 4)^2 + (y - 1)^2 = 58$

B $(x + 4)^2 + (y + 1)^2 = 58$

C $(x + 4)^2 + (y - 1)^2 = 58 + 1$

D 11

Classified Practice

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Choose the correct option.**

A 12

B $(x - 2)^2 + (y + 1)^2 = 5$ C $(x + 2)^2 + (y + 1)^2 = 5$ D $(x - 2)^2 + (y + 1)^2 = 5 + 1$

Q038 MCQ**Find the equation of the circle with centre the origin through (4,-3):****Choose the correct option.**

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 12

D 13

Classified Practice

Circles

Q039 **Written****Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**

Q040 Written**Find the equation of the circle with diameter endpoints $(3,4)$, $(5,-2)$.**

Classified Practice

Circles

Q041 **Written****Find the equation of the circle with diameter endpoints $(4,7)$, $(-2,-1)$.**

Q042 Written**Find the equation of the circle with diameter endpoints $(4,-2)$, $(-6,-2)$.**

Classified Practice

Circles

Q043 **Written****Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**

Q044 Written**Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**

Classified Practice

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Choose the correct option.**

A $(x - 2)^2 + (y + 1)^2 = 17$

B $(x + 2)^2 + (y + 1)^2 = 17$

C $(x - 2)^2 + (y + 1)^2 = 17 + 1$

D 5

Q046 MCQ**Find the equation of the circle with diameter endpoints (3,4), (5,-2):****Choose the correct option.**

A $(x - 4)^2 + (y - 1)^2 = 10 + 1$

B 8

C $(x - 4)^2 + (y - 1)^2 = 10$

D $(x + 4)^2 + (y - 1)^2 = 10$

Classified Practice

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Choose the correct option.**

A $(x - 1)^2 + (y - 3)^2 = 25 + 1$

B 14

C $(x - 1)^2 + (y - 3)^2 = 25$

D $(x + 1)^2 + (y - 3)^2 = 25$

Q048 MCQ**Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):****Choose the correct option.**

A $(x + 1)^2 + (y + 2)^2 = 25$

B $(x + 1)^2 + (y + 2)^2 = 25 + 1$

C 14**D** 15

Classified Practice

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Choose the correct option.**

A 16

B $(x + 2)^2 + (y - 1)^2 = 2$ C $(x + 2)^2 + (y + 1)^2 = 2$ D $(x + 2)^2 + (y - 1)^2 = 2 + 1$

Q050 MCQ**Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:****Choose the correct option.**

A $x^2 + (y - 2)^2 = 25 + 1$

B 17

C $x^2 + (y - 2)^2 = 25$

D $x^2 + (y + 2)^2 = 25$

Classified Practice

Circles

Q051 **Written****Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**

Q052 Written**Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**

Classified Practice

Circles

Q053 Written

Find the equation of the circle with through $(0,0)$, $(-1,1)$, $(2,4)$.

Q054 Written**Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$.**

Classified Practice

Circles

Q055 **Written****Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$.**

Q056 MCQ

Find the equation of the circle passing through (1,1), (2,-1), and (3,2):**Choose the correct option.**

A $x^2 + y^2 - 5x - y + 4 = 0$

B $x^2 + y^2 + 5x - y + 4 = 0$

C $x^2 + y^2 - 5x - y + 4 = 0 + 1$

D 6

Classified Practice

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Choose the correct option.**

A 9

B $x^2 + y^2 - 3x - 7y + 6 = 0$

C $x^2 + y^2 + 3x - 7y + 6 = 0$

D $x^2 + y^2 - 3x - 7y + 6 = 0 + 1$

Q058 MCQ**Find the equation of the circle with through (0,0), (-1,1), (2,4):****Choose the correct option.**

A $x^2 + y^2 + 2x - 4y = 0$

B $x^2 + y^2 - 2x - 4y = 0 + 1$

C 18

D $x^2 + y^2 - 2x - 4y = 0$

Classified Practice

Circles

Q059 MCQ

Find the equation of the circle with through (-4,6), (0,8), (3,-1):**Choose the correct option.**

A $x^2 + y^2 + 6y - 16 = 0$

B $x^2 + y^2 - 6y - 16 = 0 + 1$

C 19

D $x^2 + y^2 - 6y - 16 = 0$

Q060 MCQ**Find the equation of the circle with through (1,1), (5,-1), (-3,-7):****Choose the correct option.**

A $x^2 + y^2 + 2x + 8y - 8 = 0$

B $x^2 + y^2 - 2x + 8y - 8 = 0 + 1$

C 20

D $x^2 + y^2 - 2x + 8y - 8 = 0$

Classified Practice

Circles

Q061 **Challenge**

Find the circle through $(0,2)$, $(2,0)$, and $(4,6)$, giving its centre and radius.

Classified Practice

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Choose the correct option.

A $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$

B $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} + 1$

C $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10} - 1$

D 0

Classified Practice

Circles

Q063 **Written****Find the circle with centre $(4, -2)$ tangent to the x-axis.**

Q064 Written**Find the circle with centre $(-2,3)$ tangent to the y -axis.**

Classified Practice

Circles

Q065 **Written****Find the circle with centre $(-1,4)$ tangent to x-axis.**

Q066 Written**Find the circle with centre (3,-4) tangent to x-axis.**

Classified Practice

Circles

Q067 **Written****Find the circle with centre (5,-4) tangent to x-axis.**

Q068 Written**Find the circle with centre $(-1,6)$ tangent to y -axis.**

Classified Practice

Circles

Q069 **Written****Find the circle with centre $(2, -5)$ tangent to y -axis.**

Q070 Written**Find the circle with centre $(-3, -4)$ tangent to y -axis.**

Classified Practice

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Choose the correct option.**

A radius zero

B $(x - 4)^2 + (y + 2)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Choose the correct option.**

- A radius zero
- B $(x + 2)^2 + (y - 3)^2 = 4$
- C no real circle
- D centre on the wrong locus

Classified Practice

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Choose the correct option.**

A radius zero

B $(x + 1)^2 + (y - 4)^2 = 16$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 3)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 5)^2 + (y + 4)^2 = 16$

Classified Practice

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x + 1)^2 + (y - 6)^2 = 1$

Classified Practice

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Choose the correct option.**

A radius zero

B $(x - 2)^2 + (y + 5)^2 = 4$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q078 MCQ

Find the circle with centre $(-3, -4)$ tangent to y-axis:**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x + 3)^2 + (y + 4)^2 = 9$
- D no real circle

Classified Practice

Circles

Q079 **Written**

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.

Q080 Written**Find the circles through (1,2) tangent to both axes.**

Classified Practice

Circles

Q081 **Written****Find the circle with through $(-1,-2)$ tangent to both axes.**

Q082 Written**Find the circle with through $(-2,-1)$ tangent to both axes.**

Classified Practice

Circles

Q083 Written

Find the circle with through $(8,-1)$ tangent to both axes.

Classified Practice

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Choose the correct option.**

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$

Classified Practice

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Choose the correct option.**

A centre on the wrong locus

B radius zero

C $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

D no real circle

Classified Practice

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Choose the correct option.**

A radius zero

B $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$

C no real circle

D centre on the wrong locus

Classified Practice

Circles

Q089 **Written**

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

Q090 Written**Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).**

Classified Practice

Circles

Q091 **Written****Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).**

Q092 Written**Find the circle with centre on $y = 2x - 6$ through $(3,7)$ and $(-1,3)$.**

Classified Practice

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 1)^2 + (y + 1)^2 = 25$

Classified Practice

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):

Choose the correct option.

- A centre on the wrong locus
- B radius zero
- C $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
- D no real circle

Classified Practice

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Choose the correct option.**

- A centre on the wrong locus
- B radius zero
- C $(x - 2)^2 + (y - 3)^2 = 2$
- D no real circle

Classified Practice

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Choose the correct option.**

- A no real circle
- B centre on the wrong locus
- C radius zero
- D $(x - 4)^2 + (y - 2)^2 = 26$

Classified Practice

Circles

Q097 **Challenge**

A circle centre is constrained to the x-axis and passes through $P1(-2,2)$, $P2(1,1)$. Find its standard equation, centre and radius.

Classified Practice

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Choose the correct option.

A $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} - 1$

B 0

C $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

D $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} + 1$

Classified Practice

Circles

Q099 Written

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Classified Practice

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

Classified Practice

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:

Choose the correct option.

- A all real parameter values
- B radius squared is negative
- C $k < -1$ or $k > 3$
- D radius squared is zero

Classified Practice

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -\sqrt{6}$ or $k > \sqrt{6}$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Choose the correct option.**

- A all real parameter values
- B radius squared is negative
- C $k < 5/4$
- D radius squared is zero

Classified Practice

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is negative
- B $k < -1/5$ or $k > 3$
- C radius squared is zero
- D all real parameter values

Classified Practice

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Choose the correct option.

- A radius squared is zero
- B all real parameter values
- C radius squared is negative
- D $-1 < k < 4$

Classified Practice

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

Classified Practice

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Choose the correct option.

A 0

B $0 < k < 4$

C $0 < k < 4 + 1$

D $0 < k < 4 - 1$

Classified Practice

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Classified Practice

Circles

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

Classified Practice

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Classified Practice

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C $(1, -1)$ and $(-1, 3)$
- D tangent

Classified Practice

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D (0, 2) and (2, 0)

Classified Practice

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C $(3, -1)$
- D tangent

Classified Practice

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $(-2, 1)$

Classified Practice

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Choose the correct option.

- A two intersection points
- B $(-2, 1)$ and $(4, -1)$
- C tangent
- D no real intersection

Classified Practice

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

Classified Practice

Circles

Q123 Written

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Q124 Written**Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.**

Classified Practice

Circles

Q125 Written

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

Classified Practice

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Choose the correct option.**

- A tangent
- B no real intersection
- C two intersection points
- D 0 points

Classified Practice

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Choose the correct option.**

A 2points : (1, -3) and (3, 1)

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Choose the correct option.**

A no real intersection

B two intersection points

C 2 points : $\left(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2\right)$

D tangent

Classified Practice

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Choose the correct option.**

- A no real intersection
- B two intersection points
- C 0points
- D tangent

Classified Practice

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 1 point : $(-2, 0)$

Classified Practice

Circles

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Classified Practice

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Classified Practice

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Classified Practice

Circles

Q137 **Written**

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Classified Practice

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Classified Practice

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $-5 < k < 5$

Classified Practice

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{5}$

C tangent

D no real intersection

Classified Practice

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Choose the correct option.

A two intersection points

B $m = \pm\sqrt{3}$

C tangent

D no real intersection

Classified Practice

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Choose the correct option.

A two intersection points

B $0 < r < \sqrt{10}$

C tangent

D no real intersection

Classified Practice

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $m = \pm 4$
- D tangent

Classified Practice

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Choose the correct option.

A $n < -10$ or $n > 10$

B tangent

C no real intersection

D two intersection points

Classified Practice

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Choose the correct option.

- A no real intersection
- B two intersection points
- C $-5 \leq m \leq 5$
- D tangent

Classified Practice

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Choose the correct option.

A two intersection points

B $r = \sqrt{2}$

C tangent

D no real intersection

Classified Practice

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D $r > \sqrt{17}$

Classified Practice

Circles

Q149 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Classified Practice

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Choose the correct option.

A $0 < r < 2 + 1$

B $0 < r < 2 - 1$

C 0

D $0 < r < 2$

Classified Practice

Circles

Q151 **Written****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**

Q152 Written

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.

Classified Practice

Circles

Q153 **Written****Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.**

Q154 Written**Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.**

Classified Practice

Circles

Q155 Written

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.

Q156 MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

Choose the correct option.

A $\sqrt{14}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C 4

D $2\sqrt{r^2 + d^2}$

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $2\sqrt{5}$

Classified Practice

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:**Choose the correct option.**

A $2\sqrt{r^2 - d^2} + 1$

B $3\sqrt{2}$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Q160 MCQ**Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:****Choose the correct option.**

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D 2

Classified Practice

Circles

Q161 **Written**

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Classified Practice

Circles

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Classified Practice

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Choose the correct option.

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C $k = \pm 2$

D $2\sqrt{r^2 + d^2}$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $k = \pm\sqrt{31}$

Classified Practice

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Choose the correct option.

A $k = \pm 2$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Choose the correct option.

A $2\sqrt{r^2 - d^2} + 1$

B $k = 0$ or $k = -2$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

Classified Practice

Circles

Q169 **Challenge**

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Classified Practice

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Choose the correct option.

A $k = -1$ or $k = 7$

B $k = -1$ or $k = 7 + 1$

C $k = -1$ or $k = 7 - 1$

D 0

Classified Practice

Circles

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).

Q172 Written**Find the tangent to $x^2 + y^2 = 34$ at (5,3).**

Classified Practice

Circles

Q173 Written

Find the tangents $x^2 + y^2 = 10$ at (3,-1).

Q174 Written**Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$.**

Classified Practice

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.

Classified Practice

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Choose the correct option.**

A $4x + 3y = 25$

B $4x - 3y = 25$

C $4x + 3y = 25 + 1$

D the radius line

Classified Practice

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Choose the correct option.**

- A $5x + 3y = 34 + 1$
 - B the radius line
 - C $5x + 3y = 34$
 - D $5x - 3y = 34$
-
-
-

Classified Practice

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Choose the correct option.**

A $3x - y = 10$

B $3x - y = 10 + 1$

C the radius line

D 8

Classified Practice

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Choose the correct option.**

A the radius line

B 9

C $-5x - 2y = 29$ D $-5x - 2y = 29 + 1$

Classified Practice

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Choose the correct option.**

A $-2x + y = 5$

B $-2x - y = 5$

C $-2x + y = 5 + 1$

D the radius line

Classified Practice

Circles

Q181 **Written****Find the tangents from (1,3) to $x^2 + y^2 = 5$.**

Q182 Written**Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$.**

Classified Practice

Circles

Q183 Written

Find the tangents from (-1,3) to $x^2 + y^2 = 5$.

Q184 Written**Find the tangents from (2,1) to $x^2 + y^2 = 1$.**

Classified Practice

Circles

Q185 **Written****Find the tangents from (2,6) to $x^2 + y^2 = 20$.**

Classified Practice

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $2x - y = 5$ and $-x + 2y = 5$
- B $2x + y = 5$ and $-x + 2y = 5 + 1$
- C the radius line
- D $2x + y = 5$ and $-x + 2y = 5$

Classified Practice

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Choose the correct option.**

- A $-3x + y = 10$ and $x + 3y = 10$
- B $-3x - y = 10$ and $x + 3y = 10$
- C $-3x + y = 10$ and $x + 3y = 10 + 1$
- D the radius line

Classified Practice

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Choose the correct option.**

- A $-2x - y = 5$ and $x + 2y = 5$
- B $-2x + y = 5$ and $x + 2y = 5 + 1$
- C the radius line
- D $-2x + y = 5$ and $x + 2y = 5$

Classified Practice

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Choose the correct option.**

A 12

B $y = 1$ and $4x - 3y = 5$ C $y = 1$ and $4x - 3y = 5 + 1$

D the radius line

Classified Practice

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Choose the correct option.**

A $-x + 2y = 10$ and $2x + y = 10 + 1$

B the radius line

C $-x + 2y = 10$ and $2x + y = 10$

D $-x - 2y = 10$ and $2x + y = 10$

Classified Practice

Circles

Q191 **Challenge**

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Classified Practice

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Choose the correct option.

A 0

B $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

C $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) + 1$

D $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) - 1$

Classified Practice

Circles

Q193 **Written****Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$.**

Q194 Written**Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E.**

Classified Practice

Circles

Q195 **Written****Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E.**

Q196 Written**Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E.**

Classified Practice

Circles

Q197 **Written****Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.**

Q198 Written**Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.**

Classified Practice

Circles

Q199 Written

Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

Classified Practice

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C intersect at 2 points
- D separate

Classified Practice

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch externally

Classified Practice

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Choose the correct option.

- A C : touch internally
- B separate
- C internally tangent
- D intersecting at two points

Classified Practice

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D touch internally

Classified Practice

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Choose the correct option.

- A intersecting at two points
- B E : intersect at 2 points
- C separate
- D internally tangent

Classified Practice

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Choose the correct option.

- A internally tangent
- B intersecting at two points
- C D : one is inside the other
- D separate

Classified Practice

Circles

Q207 Written

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.

Q208 Written**Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.**

Classified Practice

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.

Classified Practice

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:**Choose the correct option.**

A $(x - 3)^2 + (y - 1)^2 = 45$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D $(x + 3)^2 + (y + 9)^2 = 16$

Classified Practice

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Choose the correct option.**

A $(x - 3)^2 + (y + 9)^2 = 144$

B separate

C internally tangent

D intersecting at two points

Classified Practice

Circles

Q213 **Challenge**

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Classified Practice

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Choose the correct option.

A $(x + 1)^2 + (y + 2)^2 = 9 + 1$

B $(x + 1)^2 + (y + 2)^2 = 9 - 1$

C 0

D $(x + 1)^2 + (y + 2)^2 = 9$

Classified Practice

Circles

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

Classified Practice

Circles

Q217 **Written**

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

Classified Practice

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

Choose the correct option.

A $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

B $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4 + 1$)

C the radical axis only

D 4

Classified Practice

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

Choose the correct option.

- A $x + y - 1 = 0$
- B the radical axis only
- C $x + y - 1 = 0$
- D $x - y - 1 = 0$

Classified Practice

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

Choose the correct option.

A 8

B $12x - 6y - 7 = 0$

C $12x - 6y - 7 = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:

Choose the correct option.

A $x = 1/2 + 1$

B the radical axis only

C 9

D $x = 1/2$

Classified Practice

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

Q224 Written**Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**

Classified Practice

Circles

Q225 Written

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

Classified Practice

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Choose the correct option.**

A 5

B $(-3, -1)$ and $(-1, 3)$ C $(-3, -1)$ and $(-1, 3) + 1$

D the radical axis only

Classified Practice

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:

Choose the correct option.

A the radical axis only

B 7

C (0, 2) and $(8/5, -6/5)$

D (0, 2) and $(8/5, -6/5) + 1$

Classified Practice

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Choose the correct option.**

A 10

B (1, 2) and (2, 1)

C (1, 2) and (2, 1) + 1

D the radical axis only

Classified Practice

Circles

Q229 **Challenge**

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Classified Practice

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Choose the correct option.

A $(-4, 3)$ and $(3, -4) - 1$

B 0

C $(-4, 3)$ and $(3, -4)$

D $(-4, 3)$ and $(3, -4) + 1$

Classified Practice

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Classified Practice

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$.

Classified Practice

Circles

Q235 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Classified Practice

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Choose the correct option.

A $x^2 - y^2 + 2x + 6y - 15 = 0$

B $x^2 + y^2 + 2x + 6y - 15 = 0 + 1$

C the radical axis only

D $x^2 + y^2 + 2x + 6y - 15 = 0$

Classified Practice

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Choose the correct option.

A $x^2 + y^2 + 4x + 2y = 0$

B $x^2 - y^2 + 4x + 2y = 0$

C $x^2 + y^2 + 4x + 2y = 0 + 1$

D the radical axis only

Classified Practice

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Choose the correct option.

- A $x^2 - y^2 + 10x + 20y - 39 = 0$
- B $x^2 + y^2 + 10x + 20y - 39 = 0 + 1$
- C the radical axis only
- D $x^2 + y^2 + 10x + 20y - 39 = 0$

Classified Practice

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3,2)$:

Choose the correct option.

A $x^2 + y^2 + 24x + 32y - 5 = 0 + 1$

B the radical axis only

C $x^2 + y^2 + 24x + 32y - 5 = 0$

D $x^2 - y^2 + 24x + 32y - 5 = 0$

Classified Practice

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Choose the correct option.

A the radical axis only

B $x^2 + y^2 - 10x - 6y + 24 = 0$

C $x^2 - y^2 - 10x - 6y + 24 = 0$

D $x^2 + y^2 - 10x - 6y + 24 = 0 + 1$

Classified Practice

Circles

Q241 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Classified Practice

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Choose the correct option.

- A $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$
- B $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 + 1$
- C $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 - 1$
- D 0

Classified Practice

Circles

Quiz MCQ 1 Concept Quiz · MCQ

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Choose the correct option.

- A centre $(-2, -3)$, radius 5
- B centre $(2, -3)$, radius 5
- C centre $(2, 3)$, radius 5
- D centre $(2, -3)$, radius 10

Classified Practice

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Choose the correct option.

- A tangent
- B no real intersection
- C two intersection points
- D 0

Classified Practice

Circles

Quiz MCQ 3 **Concept Quiz · MCQ**

Determine the chord length for radius r and centre-to-chord distance d :

Choose the correct option.

A $2\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

Classified Practice

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Choose the correct option.

- A separate
- B internally tangent
- C intersecting at two points
- D externally tangent

Classified Practice

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre $(1,-2)$ passing through $(4,2)$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Classified Practice

Circles

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$	Q027 $(x - 2)^2 + (y - 3)^2 = 13$	Q053 $x^2 + y^2 - 2x - 4y = 0$	Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$
Q002 $(x + 3)^2 + (y - 1)^2 = 4$	Q028 $(x + 3)^2 + (y - 1)^2 = 58$	Q054 $x^2 + y^2 - 6y - 16 = 0$	Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$
Q003 $x^2 + y^2 = 16$	Q029 $(x - 1)^2 + (y + 3)^2 = 25$	Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$	Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q004 $(x - 3)^2 + (y - 4)^2 = 36$	Q030 $(x + 4)^2 + (y - 1)^2 = 58$	Q056 A	Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q005 $(x + 2)^2 + (y - 6)^2 = 12$	Q031 $(x - 2)^2 + (y + 1)^2 = 5$	Q057 B	Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$
Q006 $x^2 + y^2 = 25$	Q032 $x^2 + y^2 = 25$	Q058 D	Q084 C
Q007 A	Q033 C	Q059 D	Q085 D
Q008 A	Q034 A	Q060 D	Q086 C
Q009 D	Q035 D	Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$	Q087 C
Q010 B	Q036 A	Q062 A	Q088 B
Q011 B	Q037 B	Q063 $(x - 4)^2 + (y + 2)^2 = 4$	Q089 $(x - 1)^2 + (y + 1)^2 = 25$
Q012 A	Q038 A	Q064 $(x + 2)^2 + (y - 3)^2 = 4$	Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
Q013 circle; centre (3, -1), radius 2	Q039 $(x - 2)^2 + (y + 1)^2 = 17$	Q065 $(x + 1)^2 + (y - 4)^2 = 16$	Q091 $(x - 2)^2 + (y - 3)^2 = 2$
Q014 circle; centre (2, 3), radius 3	Q040 $(x - 4)^2 + (y - 1)^2 = 10$	Q066 $(x - 3)^2 + (y + 4)^2 = 16$	Q092 $(x - 4)^2 + (y - 2)^2 = 26$
Q015 circle; centre (2, -4), radius 4	Q041 $(x - 1)^2 + (y - 3)^2 = 25$	Q067 $(x - 5)^2 + (y + 4)^2 = 16$	Q093 D
Q016 circle; centre (2, 0), radius 2	Q042 $(x + 1)^2 + (y + 2)^2 = 25$	Q068 $(x + 1)^2 + (y - 6)^2 = 1$	Q094 C
Q017 circle; centre (-1, 2), radius 5	Q043 $(x + 2)^2 + (y - 1)^2 = 2$	Q069 $(x - 2)^2 + (y + 5)^2 = 4$	Q095 C
Q018 circle; centre (-2, 3), radius $\sqrt{13}$	Q044 $x^2 + (y - 2)^2 = 25$	Q070 $(x + 3)^2 + (y + 4)^2 = 9$	Q096 D
Q019 A	Q045 A	Q071 B	Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$
Q020 C	Q046 C	Q072 B	Q098 C
Q021 D	Q047 C	Q073 B	Q099 $k < -1$ or $k > 3$
Q022 C	Q048 A	Q074 D	Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$
Q023 C	Q049 B	Q075 D	Q101 $k < 5/4$
Q024 C	Q050 C	Q076 D	Q102 $k < -1/5$ or $k > 3$
Q025 centre (5, 3), radius $^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$	Q051 $x^2 + y^2 - 5x - y + 4 = 0$	Q077 B	Q103 $-1 < k < 4$
Q026 B	Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$	Q078 C	Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 (1, -1) and (-1, 3)	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 (0, 2) and (2, 0)	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 (3, -1)	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 (-2, 1)	Q146 C	Q178 A	Q210 A
Q115 (-2, 1) and (4, -1)	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : (1, -3) and (3, 1)	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : (-2, 0)	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at (-2, 4); $2x + y = 10$ at (4, 0)	Q223 (-3, -1) and (-1, 3)
Q128 C	Q160 D	Q192 B	Q224 (0, 2) and (8/5, -6/5)
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 (1, 2) and (2, 1)
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 (-4, 3) and (3, -4)
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q238 D**Quiz MCQ 1** B**Quiz WRITTEN 6** $(4, 3), (-4, 3)$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q239 C**Quiz MCQ 2** D**Quiz WRITTEN 7** $3x + 4y = 25$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q240 B**Quiz MCQ 3** A**Quiz WRITTEN 8** $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$ **Q236** D

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Quiz MCQ 4 D**Q237** A**Q242** A**Quiz WRITTEN 5** $(x - 1)^2 + (y + 2)^2 = 25$